

Department of Computer and Software Engineering
SE: Machine Learning

Course Instructor: Dr. Ahmed Raza	Date: 19-February-2026
Day: Thursday	Semester: 6th

Lab 2: The Geometry of Overfitting - Polynomial Regression & Ridge

Lab Title	CLO	BT	PLO	Weightage
The Geometry of Overfitting: Polynomial Regression & Ridge				

Name	Roll Number	Report /10	Viva /5	Total /15
Zunaira Abdul Aziz	BSSE23058			

Checked on: _____

Signature: _____

1 Learning Outcomes

After completing this lab, the student will be able to:

- Construct high-degree polynomial feature matrices.
- Implement the Normal Equation using matrix inversion.
- Observe numerical instability and coefficient explosion.
- Implement Ridge Regression from scratch.
- Analyze how regularization stabilizes learning.

2 Equipment Required

- PC with Python 3.x installed (Anaconda recommended).
- IDE (Jupyter Notebook, Spyder, VS Code, or PyCharm).
- NumPy, Pandas, and Matplotlib libraries.

3 Theory and Background

High-capacity models can perfectly fit small datasets but often generalize poorly. Polynomial regression of high degree increases model flexibility but may lead to overfitting and numerical instability due to ill-conditioned matrices.

Ridge Regression introduces L2 regularization to control model complexity:

$$w = (X^T X + \lambda I)^{-1} X^T y \quad (1)$$

This shifts eigenvalues of $X^T X$, stabilizes inversion, and reduces coefficient magnitude.

4 Part A — The Overfitting Demo

Dataset: $y = \sin(2\pi x) + \epsilon$ (10 data points, fixed random seed).

Task A1: Polynomial Feature Matrix (2 Marks)

```
def build_polynomial_features(x, degree):
    X_poly = np.ones((len(x), 1))
    for d in range(1, degree + 1):
        X_poly = np.hstack((X_poly, x**d))
    return X_poly
```

```
x, y = generate_dataset(n=10)
X = build_polynomial_features(x, 20)
print(f"Matrix shape: {X.shape}")
```

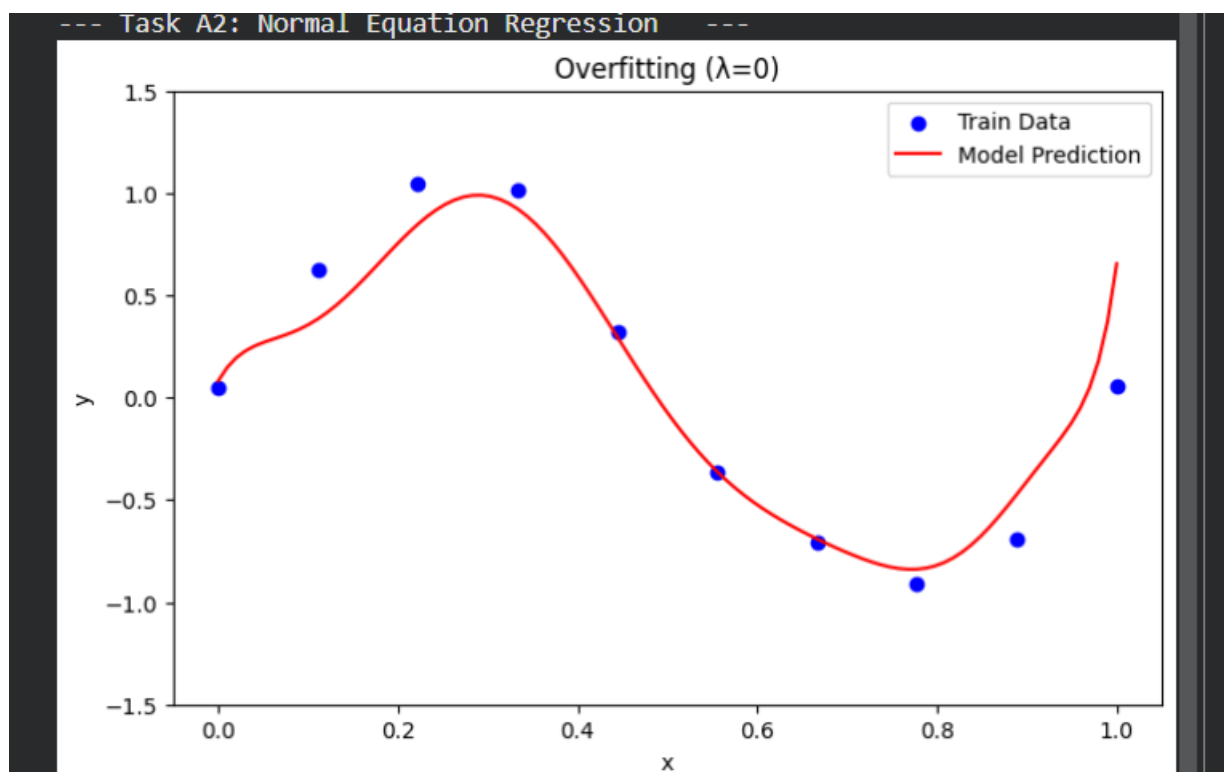
```
--- TASK A1: Polynomial Feature Matrix ---
Matrix shape: (10, 21)
```

Task A2: Normal Equation Regression (3 Marks)

```
def normal_equation(X, y):
    return np.linalg.inv(X.T @ X) @ X.T @ y

w = normal_equation(X, y)

x_test = np.linspace(0, 1, 100).reshape(-1, 1)
X_test = build_polynomial_features(x_test, 20)
y_pred = X_test @ w
plot_results(x, y, x_test, y_pred, title="20-Degree Polynomial
(Overfitting)")
```



Task A3: Coefficient Explosion & Stability (2 Marks)

```
w = normal_equation(X, y)
cond_num = compute_condition_number(X)

print(f"Weight Norm: {np.linalg.norm(w):.2e}")
print(f"Condition Number: {cond_num:.2e}")
```

```
--- Task A3: Coefficient Explosion & Stability ---
Weight Norm: 1.26e+04
Condition Number: 1.23e+18
```

Coefficient Explosion: Because the degree (20) exceeds the number of points (10), the model overfits by using massive weights to force the curve through every noisy data point.

Numerical Instability: The matrix ($X^T X$) becomes ill-conditioned (shown by a huge condition number), making the inversion process unstable and leading to unreliable, extreme coefficients.

5 Part B — Ridge Regression

Task B1: Ridge Closed-Form Implementation (3 Marks)

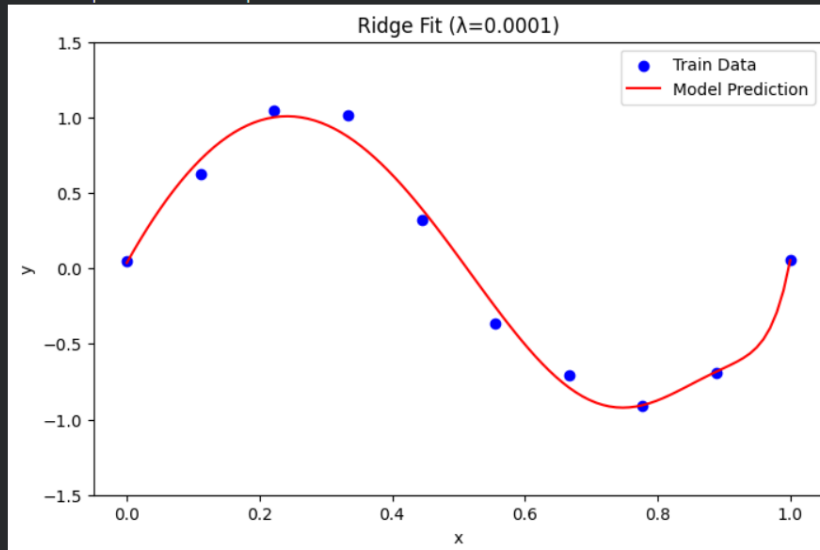
```
def ridge_regression(X, y, lambda_):
    n_features = X.shape[1]
    I = np.eye(n_features)
    I[0, 0] = 0
    w = np.linalg.inv(X.T @ X + lambda_ * I) @ X.T @ y
    return w
```

Task B2: Regularization Study (3 Marks)

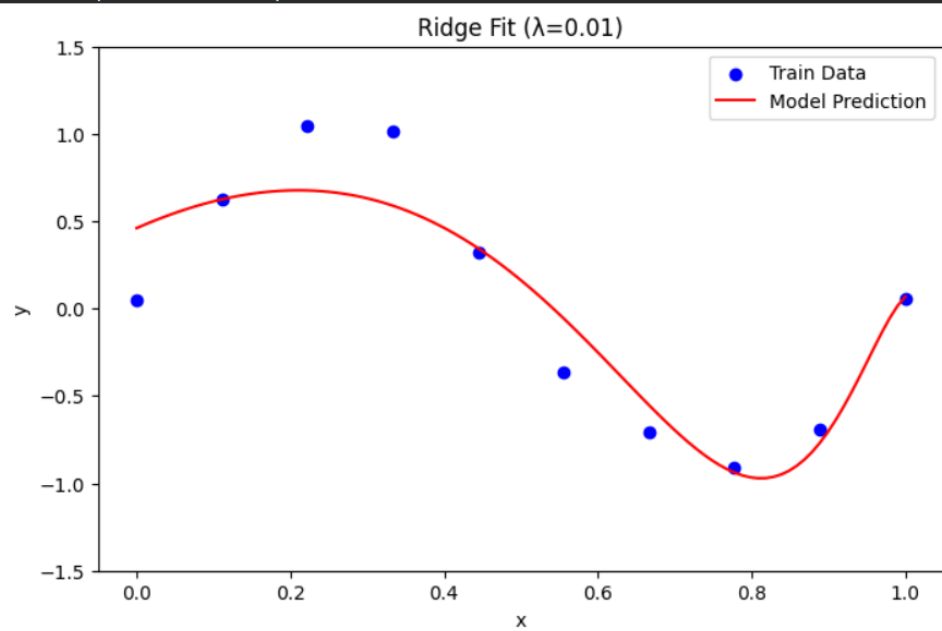
Test $\lambda \in \{0, 10^{-4}, 10^{-2}, 10^{-1}, 1\}$.

--- TASK B: Ridge Study ---

...
 $\lambda=0$ | Norm: 1.26e+04 | Condition: 1.23e+18
 $\lambda=0.0001$ | Norm: 2.22e+01 | Condition: 2.81e+05

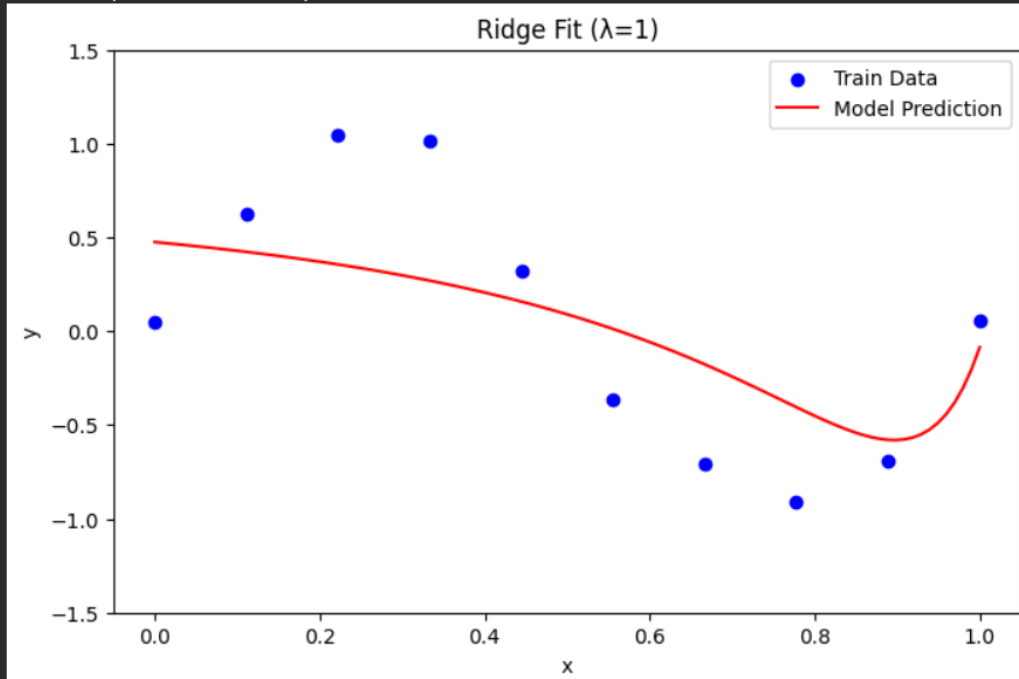


...
 $\lambda=0.01$ | Norm: 5.70e+00 | Condition: 2.81e+03



...
 $\lambda=0.1$ | Norm: 1.70e+00 | Condition: 2.81e+02

$\lambda=0.1$	Norm: 1.79e+00	Condition: 2.82e+02
$\lambda=1$	Norm: 9.64e-01	Condition: 2.89e+01



Task B3: Stability Analysis (2 Marks)

Condition Number & Eigenvalue Shift: Regularization $\lambda > 0$ adds λ to the eigenvalues of $(X^T X)$, lifting them away from zero. This reduces the condition number drastically (from $\approx 10^{18}$ to 10^3), transforming an ill-conditioned, unstable matrix into a stable, invertible one.

Geometric Meaning: Geometrically, Ridge Regression places a spherical constraint on the weight vector ($\|w\|^2 \leq t$). This prevents the weights from exploding to hit every noise point, forcing the model to favor a smoother, simpler curve that generalizes better.

ALL TEST CASES PASSED:

```
plot_results(x_train, y_train, x_plot, x_plot)
... All tests passed.
```

6 Reflection (3 Marks)

Reflection Box

1. Why does a 20-degree polynomial overfit 10 points?

The model has 21 parameters but only 10 points. It has too much "freedom," so it fits the noise instead of the trend.

2. What happens when $\lambda \rightarrow 0$?

The model becomes standard least squares and overfits with massive weights.

3. What happens when λ becomes large?

The penalty is too high; weights go to zero and the model underfits (becomes a flat line).

4. Does Ridge increase bias or reduce variance?

Ridge increases bias (simplifies the model) but reduces variance (makes it stable and better at predicting new data).

7 Conclusion (1 Mark)

Conclusion Box

Write a brief summary of what you learned from this lab, focusing on capacity control and regularization in machine learning.

I learned that high-capacity models (like high-degree polynomials) easily overfit by fitting noise, which causes numerical instability and large weights. This can be controlled through Ridge regularization, which uses a penalty term λ to stabilize matrix inversion and shrink coefficients, ensuring the model generalizes better to new data.

Assessment Rubrics

Method: Lab reports and instructor observation during lab sessions

Outcome Assessed:

1. Ability to conduct experiments, as well as to analyze and interpret data (P).
2. Ability to use the techniques, skills, and modern engineering tools necessary for engineering practice (P).

Performance	Exceeds expectation (4-5)	Meets expectation (3-2)	Does not meet expectation (1)	Marks
1. Realization of Experiment [a, b]	Selects relevant equipment to the experiment, develops setup diagrams of equipment connections or wiring.	Needs guidance to select relevant equipment and to develop equipment connection or wiring diagrams.	Incapable of selecting relevant equipment to conduct the experiment.	
2. Conducting Experiment [a, b]	Does proper calibration of equipment, carefully examines equipment moving parts, and ensures smooth operation.	Calibrates equipment, examines equipment moving parts, and operates the equipment with minor error.	Unable to calibrate appropriate equipment, and equipment operation is substantially wrong.	
3. Laboratory Safety Rules [a]	Respectfully and carefully observes safety rules and procedures.	Observes safety rules and procedures with minor deviation.	Disregards safety rules and procedures.	
6. Data Collection [a]	Plans data collection to achieve experimental objectives, and conducts an orderly and complete data collection.	Plans data collection to achieve experimental objectives, and collects complete data with minor error.	Does not know how to plan data collection; data collected is incomplete and contain errors.	
7. Data Analysis [a]	Accurately conducts simple computations and statistical analysis; correlates results to known theoretical values; accounts for measurement errors.	Conducts simple computations with minor error; reasonably correlates results to known theoretical values; attempts to account for errors.	Unable to conduct simple statistical analysis; no attempt to correlate or explain errors.	
8. Computer Use [a]	Uses computer to collect and analyze data effectively.	Uses computer to collect and analyze data with minor error.	Does not know how to use computer to collect and analyze data.	
Total				

Faculty:

Name: _____

Signature: _____

Date: _____